

Problem Set 2

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Problem 1.

$$\begin{aligned}i_{\$} &= 0.04, & i_{\text{€}} &= 0.035, \\E_{\text{€}/\$} &= 1.01, & F_{\text{€}/\$} &= 0.99.\end{aligned}$$

(a) Covered interest rate parity means:

$$(1 + i_{\$}) = (1 + i_{\text{€}}) \left(\frac{F_{\text{€}/\$}}{E_{\text{€}/\$}} \right).$$

Then:

$$\begin{aligned}1 + i_{\$} &= 1.04, \\(1 + i_{\text{€}}) \left(\frac{0.99}{1.01} \right) &\approx 1.014.\end{aligned}$$

Since $1.04 \neq 1.014$, covered interest rate parity is not holding.

(b) Uncovered interest rate parity means:

$$(1 + i_{\$}) = (1 + i_{\text{€}}) \left(\frac{E_{\text{€}/\$}^{\text{1yr}}}{E_{\text{€}/\$}} \right).$$

Because we don't know the expected future spot rate, uncovered interest rate parity is uncertain.

(c) Since the covered return is higher in the United States, the treasurer should invest in the United States

Problem 2.

$$\begin{aligned}E_{\$/\text{£}} &= 1.50, & F_{\$/\text{£}} &= 1.52, \\i_{\$} &= 0.020, & i_{\text{£}} &= 0.0145.\end{aligned}$$

(a) Covered interest rate parity does not hold. Since

$$(1 + i_{\text{£}}) \left(\frac{F_{\$/\text{£}}}{E_{\$/\text{£}}} \right) = (1.0145) \left(\frac{1.52}{1.50} \right) \approx 1.028 > 1.020,$$

a U.S. investor should borrow dollars and invest in pounds.

(b) Borrow \$1,500,000 at 2.0%:

$$(1.02)(1,500,000) = \$1,530,000.$$

Convert to pounds at the spot rate:

$$\pounds = \frac{1,500,000}{1.50} = 1,000,000.$$

Invest in the U.K.:

$$(1.0145)(1,000,000) = 1,014,500 \text{ } \pounds.$$

Convert back to dollars using the forward rate:

$$(1.52)(1,014,500) = \$1,542,040.$$

Arbitrage profit:

$$\$1,542,040 - \$1,530,000 = \$12,040.$$

Problem 3.

$$i_{\$} = 5.93\%, \quad i_{\text{lira}} = 70.0\%.$$

This is possible in the absence of default risk if the high Turkish interest rate is offset by an expected depreciation of the Turkish lira under uncovered interest rate parity. Therefore, the Turkish lira is expected to depreciate against the U.S. dollar over the coming year.